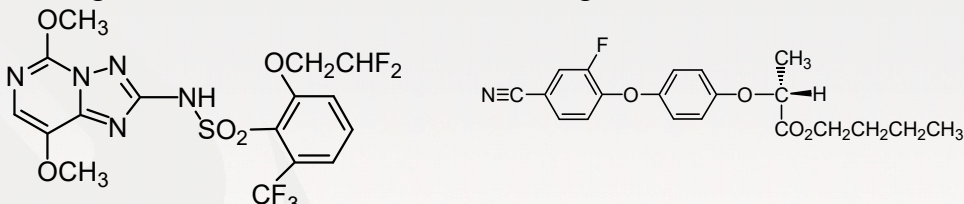




MATERIAL SAFETY DATA SHEET

1. Chemical Product Identification

Product Name:	Cyhalofop-butyl + Penoxsulam 5% + 1% WV OD	
Active ingredient:	Penoxsulam (1% W/V)	Cyhalofop-butyl (5% W/V)
Molecular Formula:	C ₁₆ H ₁₄ F ₅ N ₅ O ₅ S	C ₂₀ H ₂₀ FNO ₄
Molecular Weight:	483.37g/mol	357.38g/mol
Structural Formula:		
IUPAC	3-(2,2-difluoroethoxy)-N-(5,8-dimethoxy[1,2,4]triazolo[1,5-c]pyrimidin-2-yl)-α,α,α-trifluorotoluene-2-sulfonamide butyl (R)-2-[4-(4-cyano-2-fluorophenoxy)phenoxy]propionate	
Chemical Group:	Herbicides	Herbicide
CAS No.:	219714-96-2	122008-85-9
Manufacturer:	Shandong Binnong Technology Co., Ltd.	
Address:	No.518, Yongxin Road, Binbei Town, Binzhou, Shandong, China	

2. Hazards Identification

Classifications:	Specific Target Organ Toxicity: Repeated Category 2
	Eye Damage/Irritation: Category 2A
Signal Word (OSHA):	Warning
Hazard Statements:	Causes serious eye irritation May cause damage to organs through prolonged or repeated exposure
Hazard Symbols:	
Precautionary Statements:	Do not breathe mist, vapors, spray. Wash hands and face thoroughly after handling.

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ADD: NO.518, YONGXIN ROAD, BINBEI TOWN, BINZHOU, SHANDONG, CHINA

TEL: +86-543-5087722 FAX: +86-543-5087788

P.C.: 256600

Website: www.binnong.com

SALES COMPANY:

地址: 中国山东省滨州市黄河11路1111号

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Wear eye protection.

If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

Get medical advice if you feel unwell.

If eye irritation persists: Get medical advice.

Dispose of contents and container in accordance with local regulations.

Other Hazard Flammable hydrogen gas may be formed on contact with incompatible
Statements: metals. See "Conditions to Avoid", Section 10

3. Composition / Information on Ingredients

	Ingredient	CAS No.	Content ,g/l
Active ingredient	Penoxsulam	219714-96-2	10
Active ingredient	Cyhalofop-butyl	122008-85-9	50
Inert ingredient	Other ingredients, including water	//	Up to 1L

4. First Aid Measures

If swallowed: Call a poison control center or doctor immediately for treatment advice. Do not give any liquid to the person. Do not induce vomiting unless told to do so after calling a poison control center or doctor. Do not give anything by mouth to an unconscious person.

If in eyes: Hold eye open and rinse slowly and gently with water for 15-20 minutes. Remove contact lenses, if present, after 5 minutes, then continue rinsing eye. Call a poison control center or doctor for treatment advice.

If on skin or clothing: Take off contaminated clothing. Rinse skin immediately with plenty of water for 15-20 minutes. Call a poison control center or doctor for treatment advice.

If inhaled: Move person to fresh air. If person is not breathing, call an ambulance, then give artificial respiration, preferably mouth-to-mouth if possible. Call a poison control center or doctor for further treatment advice.

Most important symptoms/effects:

Eye irritation

Indication of immediate medical attention and special treatment needed:

There is no specific antidote if this product is ingested.

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Treat symptomatically.

5. Fire-Fighting Measures

Suitable (and unsuitable) extinguishing media:

Use dry chemical, foam or CO₂ extinguishing media. If water is used to fight fire, dike and collect runoff.

Specific Hazards:

Flammable hydrogen gas may be formed on contact with incompatible metals. See "Conditions to Avoid", Section 10.

During a fire, irritating and possibly toxic gases may be generated by thermal decomposition or combustion.

Special protective equipment and precautions for firefighters:

Wear full protective clothing and self-contained breathing apparatus. Evacuate nonessential personnel from the area to prevent human exposure to fire, smoke, fumes or products of combustion.

6. Accidental Release Measures

Personal precautions, protective equipment, and emergency procedures:

Follow exposure controls/personal protection outlined in Section 8.

Methods and materials for containment and cleaning up:

Control the spill at its source. Contain the spill to prevent from spreading or contaminating soil or from entering sewage and drainage systems or any body of water. Clean up spills immediately, observing precautions outlined in Section 8. Cover entire spill with absorbing material and place into compatible disposal container. Scrub area with hard water detergent. Pick up wash liquid with additional absorbent and place into compatible disposal container. Once all material is cleaned up and placed in a disposal container, seal container and arrange for disposition.

7. Handling and Storage

Precautions for safe handling:

Spray solutions of this product should be mixed, stored and applied using only plastic, plastic-lined steel, stainless steel or fiberglass/plastic containers. Concentrate should not be stored or maintained in long-term contact with galvanized steel, carbon steel, aluminum, brass or cast

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iron.

Store the material in a well-ventilated, secure area out of reach of children and domestic animals. Do not store food, beverages or tobacco products in the storage area. Prevent eating, drinking, tobacco use, and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

Conditions for safe storage, including any incompatibilities:

Not Applicable

8. Exposure Controls/Personal Protection

Appropriate engineering controls:

Use effective engineering controls to comply with occupational exposure limits (if applicable).

Individual protection measures:

Ingestion:

Prevent eating, drinking, tobacco usage and cosmetic application in areas where there is a potential for exposure to the material. Wash thoroughly with soap and water after handling.

Eye Contact:

Where eye contact is likely, use chemical splash goggles. Facilities storing or utilizing this material should be equipped with an eyewash facility and a safety shower.

Skin Contact:

Where contact is likely, wear chemical-resistant gloves (such as barrier laminate, butyl rubber, nitrile rubber, neoprene rubber, natural rubber, polyethylene, polyvinyl chloride [PVC] or Viton), coveralls, socks and chemical-resistant footwear.

Inhalation:

A respirator is not normally required when handling this substance. Use effective engineering controls to comply with occupational exposure limits.

9. Physical and Chemical Properties

Appearance:	off-White to Gray Liquid
Odor:	no characteristic
pH:	6.2(1% solution in deionized H ₂ O at 25°C)
Solubility in H ₂ O:	Penoxsulam: 0.0049 g/L at 19°C Cyhalofop-butyl: 0.44 mg/l at 20°C

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Vapor Pressure:

Penoxsulam:: 9.55×10^{-11} mPa at 20°C

Cyhalofop-butyl: 5.3×10^{-2} at 25°C

10. Stability and Reactivity

Reactivity: Not reactive.

Chemical stability: Stable under normal use and storage conditions.

Possibility of hazardous reactions: Will not occur.

Conditions to Avoid: Concentrate should not be stored in galvanized steel, carbon steel or aluminum. Brass and/or cast iron fittings should not be used. Spray solutions should not be mixed, stored or applied in containers other than plastic, plastic-lined steel, stainless or fiberglass/plastic.

Incompatible materials: None known.

Hazardous Decomposition Products: None known.

11. Toxicological Information

Health effects information

Likely routes of exposure: Dermal, Inhalation

Symptoms of exposure: Eye irritation

Delayed, immediate and chronic effects of exposure: Eye irritation

Acute oral toxicity: LD₅₀ for rat > 2 000 mg/kg body weight

Acute dermal toxicity: LD₅₀ for rat > 2 000 mg/kg body weight

Acute inhalation toxicity: LC₅₀ for rat > 5.0 mg/l air 4h

Acute skin irritation: non irritant to skin of rabbit

Acute eye irritation: non irritant to eye of rabbit

Skin Sensitization: weak sensitizer to skin of Guinea Pig

12. Ecological Information

Ecotoxicity Effects

Penoxsulam	Fish	LC ₅₀ for Rainbow Trout (96h) > 102 mg/l
	Blue-Green Algae	EC ₅₀ 0.49 mg/l
	Daphnia Magna	EC ₅₀ (48h) > 98.3 mg/l
	Bird	acute oral LD ₅₀ for Bobwhite Quail > 2025 mg/kg
Cyhalofop-butyl:	Birds:	LD ₅₀ for bobwhite quail (8 d) > 5625 mg/kg;

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Fish

LC₅₀ for Rainbow Trout (96h) > 0.49 mg/l

13. Disposal Considerations

Disposal:

Do not reuse product containers. Dispose of product containers, waste containers, and residues according to local health and environmental regulations.

Characteristic Waste: Not Applicable

Listed Waste: Not Applicable

14. Transport Information

DOT Classification

Ground Transport - NAFTA

Not regulated

Comments

Water Transport - International

Proper Shipping Name: Environmentally Hazardous Substance, Liquid, N.O.S., Marine Pollutant

Hazard Class: Class 3

Identification Number: UN 1993

Packing Group: PG III

Air Transport

Proper Shipping Name: Environmentally Hazardous Substance, Liquid, N.O.S.

Hazard Class: Class 3

Identification Number: UN 1993

Packing Group: PG III

15. Regulatory Information

Pesticide Registration:

This chemical is a pesticide product registered by the Environmental Protection Agency and is subject to certain labeling requirements under federal pesticide law. These requirements differ from the classification criteria and hazard information required for safety data sheets, and for workplace labels of non-pesticide chemicals. Following is the hazard information as required on the pesticide label:

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SHANDONG BINNONG TECHNOLOGY CO.,LTD

Caution: Causes moderate eye irritation. Avoid contact with eyes, skin, or clothing. Harmful if swallowed or inhaled.

16. Other Information

All information and instructions provided in this Material Safety Data Sheet (MSDS) are based on the current state of scientific and technical knowledge at the date indicated on the present MSDS and are presented in good faith and believed to be correct. This information applies to the PRODUCT AS SUCH. In case of formulations or mixes, it is necessary to ascertain that a new danger will not appear. It is the responsibility of persons in receipt of this MSDS to ensure that the information contained herein is properly read and understood by all people who use, handle, dispose or in any way come in contact with the product. If the recipient subsequently produces formulations(s) containing this product, it is the recipient's sole responsibility to ensure the transfer of all relevant information from this MSDS to their own MSDS.

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